A NOETHERIAN ALGEBRAIC SPACE WHOSE COHERENT MODULES HAVE VANISHING FIRST COHOMOLOGY IS AN AFFINE SCHEME

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Extension of the vanishing from coherent to quasi-coherent modules by filtered colimits. Same chapter spaces-cohomology.tex, labels lemma-directed-colimit-coherent and lemma-colimits (invoked at proof line 3223). They upgrade the hypothesis $H^1(X, \mathcal{F}) = 0$ for coherent $\mathcal{F}$ to the same vanishing for every quasi-coherent module.
- Serre's criterion for affineness in the vanishing situation. Same chapter spaces-cohomology.tex, label proposition-vanishing-affine (invoked at proof line 3225). Once the vanishing holds for all quasi-coherent modules, it gives that X is an affine scheme.

Source reference. Original lemma 073E: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 667–667. Navigation pointer for \ref{lemma-colimits} used inside the counted span (line 3223); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UV: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2866–2866. Navigation pointer for \ref{lemma-directed-colimit-coherent} used inside the counted span (line 3223); the argument that consumes it is retained in the span.

Source reference. Original proposition 07V6: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 3160–3160. Navigation pointer for \ref{proposition-vanishing-affine} used inside the counted span (line 3225); the argument that consumes it is retained in the span.

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Assume that for every coherent $\mathcal{O}_X$-module $\mathcal{F}$ we have $H^1(X, \mathcal{F}) = 0$. Then X is an affine scheme.*

Proof. The assumption implies that $H^1(X, \mathcal{F}) = 0$ for every quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ by Lemmas 07UV and 073E. Then X is affine by Proposition 07V6. □